CERTIFIED MATERIAL TEST REPORT
CERTIFICATE OF COMPLIANCE
CERTIFICATE # 250731
REVISION 0

Tube Forgings of America, Inc.
5200 N.W. Front Avenue
Portland, OR 97210
ISO 9001:2015 Certified

6/29/2026

Sold To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140	Ship To	DNOW LP #MI04 1701 Gover Pkwy. Mt Pleasant, MI 48858-8140
Customer ID	34217		
TFA Sales Order	331045\1-1		
Purchase Order	663013401		

TFA Part ID	00059	Heat Code	A3118
Description	4 XH LR 90	Grade	WPB
Quantity	16	Raw Material Heat	4113856
Customer Part ID	3674	Process	WPB-H
		Mill	JIANGYIN

Austenitized @: Held For: mins/min	Tempered @: Held For: mins/min	Stress Relieved @: Held For: mins/min	Medium:

Heat Analysis Chemistry

C	Mn	P	S	Si	Mo	Cr	Ni	Cu	V	Cb	B	Ti	Al	N	Co	Ca	C.E.
0.20	0.46	0.010	0.006	0.25	<0.01	0.03	0.02	0.02	<0.01								0.29

Tensile Test Results

Test	Name	Yield	Tensile	% Elong	R/A %	Gage Length	HBW	*In-Process* Test 1	Test 2	Tensile Orientation	Test Bar Type & Size
1	BASE	43366 (psi)	64976 (psi)	31.60		2"	136	135	138	LONGITUDINAL	STRIP

Charpy Impact Test Results

	Impact Energy				Lateral Expansion (Mils)			% Shear				Test Temp.	Size (mm)	Charpy Orientation
	1	2	3	Avg	1	2	3	1	2	3	Avg			

Notes/Comments

1.THESE FITTINGS WERE HOT FORMED BETWEEN 1150FAND1800F AND AIR COOLED. 2.THE HARDNESS DOES NOT EXCEED 197 BHN PART TO PART. 3. STARTING MATERIAL IS ASTM A 106 GRB SEAMLESS PIPE. 4. THIS PRODUCT IS SEAMLESS.

1) MATERIAL IS MANUFACTURED TO ASTM A234-24 GR WPB, ALSO PER ASME SEC II SA234 GR WPB-2023 EDITION. **2)** DESIGN AND DIMENSIONS TO ASME B16.9-2018. **3)** MATERIAL MEETS ANSI/NACE MR0175/ISO15156:2015 AND ANSI/ NACE MR0103/ISO17945:2015. **4)** MANUFACTURED IN AN ISO 9001REGISTERED FACILITY, CERTIFICATE #: 30248. **5)** TFA, INC. ORGANIZATIONAL STRUCTURE AND CERTIFICATION COMPLIES WITH BS EN 10204-2004 TYPE 3.1.B. **6)** TFA INC. FITTINGS SATISFY THE MATERIAL REQUIREMENT OF PRESSURE EQUIPMENT DIRECTIVE (PED) 2014/68/EU, ANNEX 1, PARAGRAPH 4.3. **7)** FITTINGS ARE MANUFACTURED IN THE USA. RAW MATERIAL IS FROM DOMESTIC AND GLOBAL SOURCES. **8)** FITTINGS WERE MANUFACTURED, SAMPLED, TESTED AND INSPECTED IN ACCORDANCE WITH THE ORDER AND SPECIFICATIONS AND WERE FOUND TO MEET THE REQUIREMENTS.

*UB = Unbroken Test

Heat: A3118 Grade: WPB

Scot Marshall, Manager Quality Assurance

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